CCNA Practice Final 1

Implementación de redes de área amplia y servicios distribuidos (TC3003B.570)

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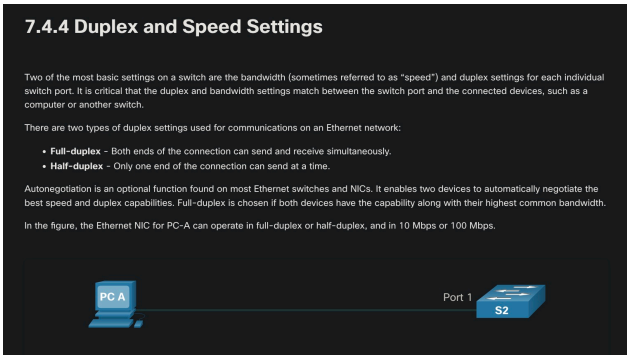
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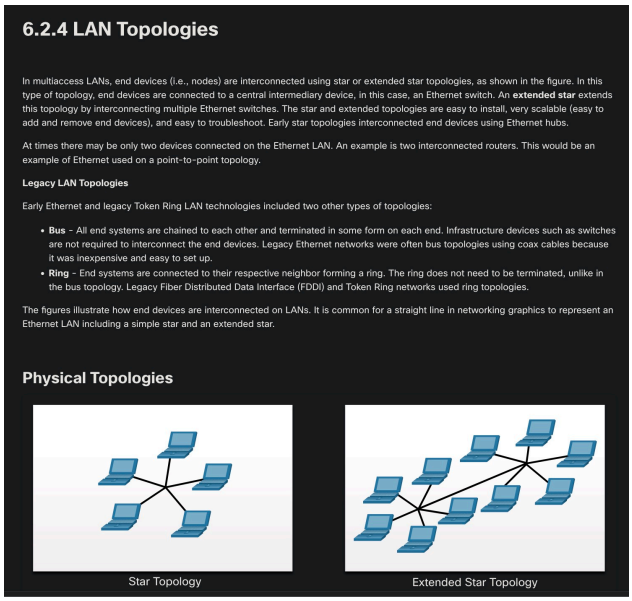
1. Which two statements describe the characteristics of fiber-optic cabling? (Choose two.)
   - Respuesta: Fiber-optic cabling does not conduct electricity.
   - Respuesta: Fiber-optic cabling is primarily used as backbone cabling.
   - Los hilos de vidrio aislante transmiten pulsos luminosos inmunes a la interferencia electromagnética.
   - Su alto ancho de banda convierte a la fibra en el medio principal para enlaces troncales.
2. Although CSMA/CD is still a feature of Ethernet, why is it no longer necessary?
   - Respuesta: the use of full-duplex capable Layer 2 switches
   - Los switches de Capa 2 operan en dúplex completo y permiten la transmisión bidireccional simultánea sin colisiones en el medio.
3. Which two acronyms represent the data link sublayers that Ethernet relies upon to operate? (Choose two.)
   - Respuesta: MAC
   - Respuesta: LLC
   - El estándar IEEE 802 divide la capa de enlace de datos en las subcapas LLC y MAC.
4. A network team is comparing topologies for connecting on a shared media. Which physical topology is an example of a hybrid topology for a LAN?
   - Respuesta: extended star
   - La topología en estrella extendida interconecta switches centrales con múltiples switches de acceso en una estructura híbrida.
5. What does a router do when it receives a Layer 2 frame over the network medium?
   - Respuesta: de-encapsulates the frame
   - El router retira el encabezado y el tráiler de Capa 2 para leer el paquete IP contenido en la trama.
6. What routing table entry has a next hop address associated with a destination network?
   - Respuesta: remote routes
   - Las rutas remotas dirigen el tráfico hacia redes no locales mediante la dirección IP del siguiente router en la ruta.
7. Which term describes a field in the IPv4 packet header that contains a unicast, multicast, or broadcast address?
   - Respuesta: destination IPv4 address
   - El campo de dirección IPv4 de destino almacena los 32 bits que identifican al receptor individual, de grupo o de difusión.
8. Refer to the exhibit. PC1 issues an ARP request because it needs to send a packet to PC3. In this scenario, what will happen next?
   - Respuesta: RT1 will send an ARP reply with its own Fa0/0 MAC address.
   - PC1 debe enviar el paquete fuera de su subred local y solicita la dirección MAC de la interfaz Fa0/0 del router RT1.
9. Refer to the exhibit. The exhibit shows a small switched network and the contents of the MAC address table of the switch. PC1 has sent a frame addressed to PC3. What will the switch do with the frame?
   - Respuesta: The switch will forward the frame to all ports except port 4.
   - Al no encontrar la dirección MAC de PC3 en su tabla, el switch inunda la trama por todos los puertos

activos excepto el puerto 4 de entrada.

10. Match the command with the device mode at which the command is entered.
    - Respuesta: service password-encryption -> R1(config)#
    - Respuesta: enable -> R1>
    - Respuesta: copy running-config startup-config -> R1#
    - Respuesta: login -> R1(config-line)#
    - Respuesta: ip address 192.168.4.4 255.255.255.0 -> R1(config-if)#
    - Cada comando de Cisco IOS pertenece a un nivel específico: usuario, privilegiado, configuración global o subconfiguración de interfaz y línea.
11. Refer to the exhibit. A network administrator is connecting a new host to the Registrar LAN. The host needs to communicate with remote networks. What IP address would be configured as the default gateway on the new host?
    - Respuesta: 192.168.235.234
    - La puerta de enlace predeterminada corresponde a la dirección IP 192.168.235.234 asignada a la interfaz gi0/1 del router en esa subred.

Anexo: Capturas del material de Cisco Introduction to Networks





6.1.3 Providing Access to Media

Each network environment that packets encounter as they travel from a local host to a remote host can have different characteristics. For example, an Ethernet LAN usually consists of many hosts contending for access on the network medium. The MAC sublayer resolves this. With serial links the access method may only consist of a direct connection between only two devices, usually two routers. Therefore, they do not require the techniques employed by the IEEE 802 MAC sublayer.

Router interfaces encapsulate the packet into the appropriate frame. A suitable media access control method is used to access each link. In any given exchange of network layer packets, there may be numerous data link layers and media transitions.

At each hop along the path, a router performs the following Layer 2 functions:

1. Accepts a frame from a medium
2. De-encapsulates the frame
3. Re-encapsulates the packet into a new frame
4. Forwards the new frame appropriate to the medium of that segment of the physical network

Press play to view the animation. The router in the figure has an Ethernet interface to connect to the LAN and a serial interface to connect to the WAN. As the router processes frames, it will use data link layer services to receive the frame from one medium, de-encapsulate it to the Layer 3 PDU, re-encapsulate the PDU into a new frame, and place the frame on the medium of the next link of the network.

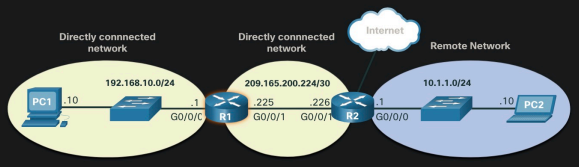
8.5.2 IP Router Routing Table

The routing table of the router contains network route entries listing all the possible known network destinations.

The routing table stores three types of route entries:

- **Directly-connected networks** - These network route entries are active router interfaces. Routers add a directly connected route when an interface is configured with an IP address and is activated. Each router interface is connected to a different network segment. In the figure, the directly-connected networks in the R1 IPv4 routing table would be 192.168.10.0/24 and 209.165.200.224/30.
- **Remote networks** - These network route entries are connected to other routers. Routers learn about remote networks either by being explicitly configured by an administrator or by exchanging route information using a dynamic routing protocol. In the figure, the remote network in the R1 IPv4 routing table would be 10.1.1.0/24.
- **Default route** - Like a host, most routers also include a default route entry, a gateway of last resort. The default route is used when there is no better (longer) match in the IP routing table. In the figure, the R1 IPv4 routing table would most likely include a default route to forward all packets to router R2.

The figure identifies the directly connected and remote networks of router R1.



R1 has two directly connected networks:

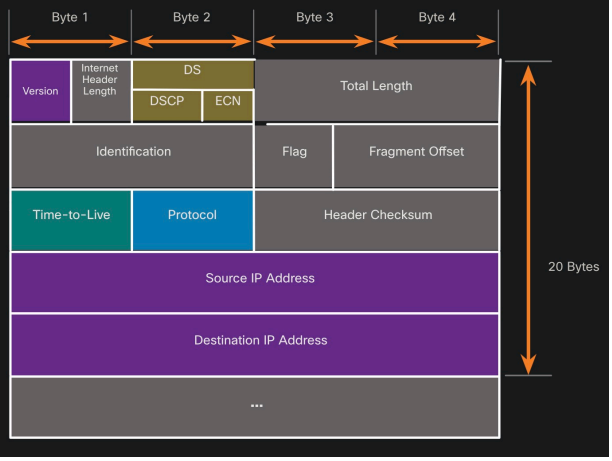
- *192.168.10.0/24*
- *209.165.200.224/30*

R1 also has remote networks (i.e. 10.1.1.0/24 and the internet) that it can learn about.

8.2.2 IPv4 Packet Header Fields

The binary values of each field identify various settings of the IP packet. Protocol header diagrams, which are read left to right, and top down, provide a visual to refer to when discussing protocol fields. The IP protocol header diagram in the figure identifies the fields of a IPv4 packet.

Fields in the IPv4 Packet Header



9.2.2 ARP Functions

When a packet is sent to the data link layer to be encapsulated into an Ethernet frame, the device refers to a table in its memory to find the MAC address that is mapped to the IPv4 address. This table is stored temporarily in RAM memory and called the ARP table or the ARP cache.

The sending device will search its ARP table for a destination IPv4 address and a corresponding MAC address.

- If the packet's destination IPv4 address is on the same network as the source IPv4 address, the device will search the ARP table for the destination IPv4 address.
- If the destination IPv4 address is on a different network than the source IPv4 address, the device will search the ARP table for the IPv4 address of the default gateway.

In both cases, the search is for an IPv4 address and a corresponding MAC address for the device.

Each entry, or row, of the ARP table binds an IPv4 address with a MAC address. We call the relationship between the two values a map. This simply means that you can locate an IPv4 address in the table and discover the corresponding MAC address. The ARP table temporarily saves (caches) the mapping for the devices on the LAN.

If the device locates the IPv4 address, its corresponding MAC address is used as the destination MAC address in the frame. If there is no entry is found, then the device sends an ARP request.

Click Play in the figure to see an animation of the ARP function.

7.3.2 Switch Learning and Forwarding

The switch dynamically builds the MAC address table by examining the source MAC address of the frames received on a port. The switch forwards frames by searching for a match between the destination MAC address in the frame and an entry in the MAC address table.

Click the Learn and Forward buttons for an illustration and explanation of this process.

Learn | Forward

Find the Destination MAC Address

If the destination MAC address is a unicast address, the switch will look for a match between the destination MAC address of the frame and an entry in its MAC address table. If the destination MAC address is in the table, it will forward the frame out the specified port. If the destination MAC address is not in the table, the switch will forward the frame out all ports except the incoming port. This is called an unknown unicast.

As shown in the figure, the switch does not have the destination MAC address in its table for PC-D, so it sends the frame out all ports except port 1.

Note: If the destination MAC address is a broadcast or a multicast, the frame is also flooded out all ports except the incoming port.

10.3.1 Default Gateway on a Host

If your local network has only one router, it will be the gateway router and all hosts and switches on your network must be configured with this information. If your local network has multiple routers, you must select one of them to be the default gateway router. This topic explains how to configure the default gateway on hosts and switches.

For an end device to communicate over the network, it must be configured with the correct IP address information, including the default gateway address. The default gateway is only used when the host wants to send a packet to a device on another network. The default gateway address is generally the router interface address attached to the local network of the host. The IP address of the host device and the router interface address must be in the same network.

For example, assume an IPv4 network topology consisting of a router interconnecting two separate LANs. G0/0/0 is connected to network 192.168.10.0, while G0/0/1 is connected to network 192.168.11.0. Each host device is configured with the appropriate default gateway address.

In this example, if PC1 sends a packet to PC2, then the default gateway is not used. Instead, PC1 addresses the packet with the IPv4 address of PC2 and forwards the packet directly to PC2 through the switch.